Packet Tracer - WAN Concepts

Answer Sheet

Part 1: Investigate Consumer WAN Technologies for Home and Mobile Devices

Step 1(a) — What are the WAN technologies in use?

The two home networks use Cable and DSL WAN technologies. A third consumer WAN technology shown in the activity is cellular networking for the smartphone.

Step 1(b) — What media is used to connect the two home networks to the ISP? What devices in the home networks are directly connected to the ISP?

The Home Cable Network uses coaxial cable to connect to the ISP/Telco through the coaxial splitter. The Home DSL Network uses a telephone/DSL line to connect to the ISP/Telco. The cable modem and DSL modem are the home devices that provide the connection to the ISP service.

Step 1(c) — What ports are available on the DSL modem and what is connected to them?

The DSL modem has a DSL/telephone-line interface connected toward the ISP/Telco network and an Ethernet interface connected to Wireless Router2.

Step 1(c) — What is the purpose of the DSL modem?

The DSL modem converts/modulates digital network data so it can travel over the telephone/DSL line, and converts incoming DSL signals back into data for the home network.

Step 1(c) — What is the type of connection between the ISP/Telco/Cable Company network and the Home Cable Network? Why is the splitter necessary?

The connection is a coaxial cable connection. The splitter is necessary because it allows the coaxial cable service to be shared between the cable modem and the TV while using the same incoming cable line.

Step 1(d) — What does the cable modem do? What connections does it have?

The cable modem converts between the cable company's network signals and Ethernet data used by the home network. It has a coaxial cable connection toward the cable service and an Ethernet connection toward the wireless router.

Step 1(d) — Which port does the cable from the cable modem connect to on the home wireless router? Where did the interface IP address come from?

The Ethernet cable from the cable modem connects to the Internet/WAN port of the home Wireless Router. The router's Internet/WAN interface receives its IP addressing automatically through DHCP from the ISP/Telco network.

Step 1(e) — What is the smartphone's IP address? Where did the IP address come from?

The smartphone IP address is 198.51.100.100. The address was assigned by DHCP from the Telco/cellular network.

Step 1(e) — What data service is the cellphone currently using (cellular data or Wi-Fi)?

The smartphone is currently using cellular data through the 3G/4G cellular network, not Wi-Fi.

Step 2: Explore the Business WAN

Step 2(a) — What different types of connections do you see in use in the Business network?

The Business WAN uses Ethernet over copper, Ethernet over fiber-optic cable, and serial WAN connections.

Step 2(b) — What types of interfaces are present on the StoreNet switch?

The StoreNet switch has Gigabit Ethernet copper interfaces and four modular ports. Three of the modular ports contain GLC-LH-SMD fiber-optic SFP modules.

Step 2(b) — Which interfaces and media are used to connect the store networks to the Business Headquarters network? Why was this done?

The store networks are connected to StoreNet using Ethernet over fiber-optic cable. Fiber was used because the stores are located at a distance from the Business Headquarters/StoreNet, so fiber is suitable for longer-distance connections.

Step 2(b) — What type of WAN service is used to connect the Business Headquarters router to the ISP?

A serial WAN connection is used to connect the Business Headquarters router to the ISP.

Part 2: Explore Connectivity

Can all hosts ping each other and the web server?

No. Hosts within the same WAN/network can communicate, and both Business WAN and Consumer WAN hosts can reach the Web Server. However, hosts in the Business WAN cannot directly ping hosts in the Consumer WAN, and Consumer WAN hosts cannot directly ping Business WAN hosts. For example, PC0 (192.168.10.2) successfully reached the Web Server (209.165.200.226), while Laptop 2 (192.168.0.100) could not reach PC0 (192.168.10.2). Laptop 2 could communicate with Laptop 1 (192.168.0.102) within the Consumer WAN.

Is this a good situation?

Yes. This is a good situation because the Business WAN and Consumer WAN should not be directly reachable from each other. Separating the networks improves security while still allowing authorized communication to the Web Server.

Key Ping Results from the Activity

Ping Test	Result
PC0 192.168.10.2 → Store 2-1 192.168.20.3	Successful
PC0 192.168.10.2 → Web Server 209.165.200.226	Successful
Laptop 2 192.168.0.100 → Laptop 1 192.168.0.102	Successful
Laptop 2 192.168.0.100 → PC0 192.168.10.2	Unsuccessful / Destination host unreachable
Laptop 2 192.168.0.100 → Web Server 209.165.200.226	Successful